Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	DNA Minicircle-Based Mechanical Reporters for Drug Discovery				
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (v), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )				
Industrial Technology Classification [1)]	Technology Field 1 (Large Category)			Technology Field 2 (Large Category)	
	Advanced Bio				
	Medium Category		Small Category	Medium Category	Small Category
	Advanced Bio				
Concept of Proposed Technology	This project proposes a DNA-based screening platform built around short double-stranded DNA minicircles (~100 base pairs) that function as mechanically stressed, topology-sensitive reporters of DNA-drug interactions. At this length scale, circular DNA is uniformly and highly bent, and torsional perturbations globally impact the molecule without dissipation. As a result, even subtle changes in DNA twist or writhe, such as those induced by intercalators or topoisomerase modulation, are amplified into measurable conformational changes. The platform converts these mechanical changes into robust optical readouts using standard fluorescence techniques. Many clinically important compounds act by altering DNA topology, yet most current screening assays rely on long plasmids or linear DNA fragments, which attenuate the effect of local structural perturbations resulting from DNA-drug interactions. In contrast, DNA minicircles exhibit strong coupling between twist, bending, and writhe. This makes them exceptionally sensitive to topology-altering				

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	interactions. By working in a well-defined mechanical regime, the platform enables direct, physics-based detection of drug action on DNA itself with unprecedented sensitivity. For the proposed technology development, minicircles will be engineered with asymmetrically positioned fluorescent probes that report conformational changes via Förster resonance energy transfer (FRET). Binding of a compound or modulation of topoisomerase activity redistributes torsional stress within the closed DNA loop, producing a detectable shift in FRET efficiency. The assay does not rely on rare topological transitions or strand crossing events, ensuring robustness and reproducibility. Measurements are compatible with standard plate readers and scalable to high-throughput formats. Compared to plasmid-based or linear DNA assays, the minicircle platform offers substantially enhanced sensitivity to topology-altering compounds while requiring minimal material. The homogeneous and topologically defined substrate reduces background variability and improves assay consistency. Extrachromosomal DNA (ecDNA), which is prevalent in many aggressive cancers, exists in circular, highly transcribed, and mechanically stressed states that are fundamentally distinct from linear chromosomal DNA. The minicircle platform mimics key physical features of ecDNA, including closed topology, enhanced torsional stress, and sensitivity to transcription-associated perturbations. As a result, the assay provides a unique opportunity to identify compounds that preferentially act on stressed circular DNA substrates. This opens a path toward discovering agents with selective activity against ecDNA-rich cancer cells.
Rationale for Joint R&D Project	The proposed collaboration is well suited to a joint R&D project because it combines complementary expertise that is difficult to assemble within a single organization. The academic side contributes expertise in DNA mechanics, minicircle construction, and quantitative interpretation of topology-driven conformational changes (See our relevant publications in the reference section), while an industry partner contributes compound libraries, assay automation, and translational expertise. Companies specializing in fluorescence reagents, DNA labeling chemistry, optical readout platforms, or microfluidic assay formats are natural collaborators for this effort. Their existing capabilities can be leveraged to optimize dye placement strategies, improve signal stability and reproducibility, and translate laboratory-scale measurements into standardized, manufacturable formats.
Key Words	1. DNA minicircle 2. Topology-sensitive conformational changes 3. Mechanics-amplified signal 4. Fluorescence/FRET-based detection 5. Cancer drug screening

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]		**Start of the Project**	Level ( )	**End of the Project**	Level ( )
R&D Objectives		The development of a topology-sensitive DNA minicircle platform for mechanism-driven screening of compounds that modulate DNA structure and enzymatic activity. The technology should be applicable to different classes of DNA-binding compounds, including topoisomerase inhibitors, intercalators, and other torsion-sensitive molecules, and for a range of assay formats including fluorescence-based detection and high-throughput screening. The platform is designed to be extensible to study extrachromosomal DNA (ecDNA) in cancer cells and to enable the identification of compounds with selective activity toward mechanically stressed circular DNA substrates.			
R&D Target Specifications		The criteria to measure success in the proposed research include: • Sensitivity to Topological Perturbations: Magnitude and reproducibility of fluorescence signal changes in response to defined torsional stress or intercalator/topoisomerase activity on the minicircle substrate. • Scalability and Throughput: Number of compounds that can be screened in parallel using the minicircle assay while maintaining signal fidelity and reproducibility. • Assay Robustness: Stability of the fluorescence readout under variations in temperature, ionic strength, buffer composition, and minicircle sequence or length. • Selectivity for Circular DNA: Ability to discriminate compounds that preferentially act on stressed circular DNA versus linear DNA or relaxed substrates. • Adaptability to Biologically Relevant Targets: Performance in detecting torsionally sensitive interactions on ecDNA-like substrates, enabling prioritization of compounds with potential cancer-cell specificity.			
R&D Plan		**3-year Milestone Outline** Year 1. Proof-of-Concept • Synthesize 100-bp DNA minicircle library with fluorescent probes. • Establish baseline FRET readouts and validate sensitivity to model topoisomerase inhibitors and intercalators. • **Deliverable**: reproducible minicircle assay demonstrating measurable topology-dependent signal changes. Year 2. Optimization and Scale-Up • Refine probe placement, sequence composition, and assay conditions for maximal signal-to-noise. • Adapt assay for moderate-throughput compound screening. • **Deliverable**: standardized, scalable assay protocol with defined metrics			

		for sensitivity, selectivity, and robustness. Year 3. ecDNA Application and Mechanistic Profiling • Apply assays to ecDNA-like circular substrates to test selectivity and stress sensitivity. • Screen small compound libraries and generate mechanistic profiles. • **Deliverable**: validated platform with cancer-relevant application, ready for commercialization or translational R&D.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Kim lab at Georgia Tech will provide core expertise in DNA mechanics, minicircle design, and quantitative analysis of torsionally stressed substrates (See our relevant publications in the reference section). Responsibilities include the design and synthesis of DNA minicircle constructs, placement of fluorescent probes for topology-sensitive readouts, and development of baseline fluorescence and FRET measurement protocols. Our team will also generate mechanistic insights into torsional stress redistribution, supervise assay validation, and provide analytical support for data interpretation. Additionally, we will document protocols, define assay performance metrics, and contribute to experimental design for mechanistic profiling of DNA-targeting compounds, ensuring reproducibility and scientific rigor.
Expected R&R of Korean SME Partner	The Korean SME partner will contribute capabilities in assay scale-up, high-throughput screening, and translational development. Responsibilities include adapting validated minicircle assays for automated or parallel screening workflows, providing access to compound libraries, optimizing assay robustness under industrial conditions, and generating standardized data suitable for decision-making and potential commercialization. The SME partner will also advise on quality control, reproducibility, and regulatory considerations, support development of assay kits or platforms, and collaborate on refinement of probe designs and workflows to maximize signal stability and screening efficiency. Finally, it will participate in co-designing experiments that evaluate selectivity toward stressed circular DNA, including ecDNA-relevant substrates.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcomes will include: 1. A validated DNA minicircle-based assay platform with reproducible, topology-sensitive fluorescence readouts suitable for high-throughput compound screening. 2. Optimized probe designs, assay workflows, and detection protocols that can be developed into commercially deployable research-use kits. 3. Demonstrated capability to discriminate compounds that preferentially act on torsionally stressed circular DNA, including substrates mimicking ecDNA in cancer cells. 4. Data and methods enabling mechanism-driven profiling of topoisomerase inhibitors, intercalators, and other DNA-targeting agents. The impact of the proposed work is to address a critical need for sensitive, mechanistically informed screening tools for DNA-targeting therapeutics. While the initial focus is on circular DNA substrates and ecDNA-relevant targets, the platform can be extended to other torsionally stressed DNA forms, enabling broader applications in cancer research, drug discovery, and fundamental studies of DNA mechanics.

6. References

(1) Lemos, T., & Kim, H. D. (2025). Visualizing Poloidal Orientation in DNA Minicircles. Preprint. https://doi.org/10.48550/arXiv.2508.15091

(2) Hart, D. J., Jeong, J., Gumbart, J. C., & Kim, H. D. (2023). Weak tension accelerates hybridization and dehybridization of short oligonucleotides. Nucleic Acids Research. https://doi.org/10.1093/nar/gkad118

(3) Moon, Y., Moon, H., Chang, J., Kim, H. D., Lee, J. H., & Lee, J. (2022). Development of a highly sensitive lateral flow strip device for nucleic acid detection using molecular beacons. Frontiers in Sensors. https://doi.org/10.3389/fsens.2022.1012775

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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